Do AI Writing Assistants Improve Productivity?

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ABSTRACT

In the last 12 months, AI-powered writing assistants and especially AI-powered writing generators have attracted attention worldwide. To best understand the potential of writing generators, it will be helpful to first understand the impact of writing assistants. As writing assistants and generators continue to proliferate, the research community should develop clearer definitions and frameworks for both categories of AI-powered tools. By creating a taxonomy of the rapidly emerging writing generators, the second In2Writing Workshop has the opportunity to influence the reception of writing generators currently under development.

KEYWORDS

AI-powered writing assistants, AI-powered writing generators, Google Smart Compose

1 INTRODUCTION

A team of researchers at the University of Mississippi has conducted human participant research to better understand the impact of writing assistants on writing products and processes. Because this research has been submitted to an upcoming conference, certain rules preserving the originality of that conference submission must be followed. However, because the In2Writing workshop is non-archival, the conference where the paper has been submitted will permit reference to a limited percentage of that submission. Therefore this position statement will outline that work and then continue to discuss how that work will apply to the goals of In2Writing, which include drafting a taxonomy for better understanding of the rapidly evolving landscape of AI writing assistants and generators.

1.1 Google Smart Compose Research Brief Project Description

We recruited 119 participants at a US public research university to write for 25 minutes in response to a prompt that solicited their opinions about the roles of luck and/or hard work in accounting for success in life. No university status was sought or required (i.e., no student status was required), no literacy ability was measured, and all participants were 18 years of age or older. We randomly enabled/disabled Google Smart Compose for each participant via uniform sampling. This resulted in roughly half (55) of the participants writing with Google Smart Compose enabled, and roughly half (59) writing without Google Smart Compose enabled. All participants wrote on Google Docs via laptop computers in a room with up to four other participants, with access to the room controlled by a research assistant. The 25 minute time limit for sample collection was monitored by the software, and the research assistant guided participants to and from laptop work stations after participants had completed the consent process. Participants were deceived as to the purpose of the study and our intent to measure the impact Google Smart Compose, though they were fully informed after the experiment.Our study was reviewed and received exempt status from our institutional IRB office.

1.2 Limited Results

We chose to collect data about (a) the writing product (the text), and (b) the writing process (the writer's actions). Although our conference submission will discuss the results in greater detail, we can offer the following limited results as represented in Table 1.

1.3 Limited Discussion

After noting that in published research on writing assistants in mobile platforms the trend for auto completion technologies was to both shorten writing samples and to suppress diversity of written expression (Arnold, Chauncey, and Gajos, 2020) [1], we sought to understand if that effect would hold when human writers engaged an open-ended writing response to a common prompt on a word processing platform. We believe that these data demonstrate at least two important findings: writing assistants, engaged as auto completion technologies in word processors in open-ended writing tasks, do not impact the amount of writing produced by human writers, nor do they suppress diversity of expression, as measured by lexical diversity, syntactic complexity, or ease of reading tools (e.g., Coh-Metrix).

These findings will need to be further contextualized in research literature, but have potentially important impact of the discussions at In2writing. First, a central claim for the newest implementation of writing assistants into word processors, such as the proposed integration of Bing search assistants into Microsoft Word, is that they will increase productivity. Our findings are that writing assistants do not automatically translate to greater productivity, at least when measured strictly in terms of word output. Second, our findings are consistent with the theory that writers are able to adjust relatively quickly to the implementation of new writing assistant technologies, as demonstrated by the effect that auto completion technology did not extend the amount of writing when present or absent for human writers pursuing open-ended writing tasks on word processors. More research is needed to track the uptake of human writers engaging the latest round of writing assistant technologies, but our preliminary findings are good news for the work of Dang, et al., who propose relatively complex AI- interactions of automatic text summarization as writing assistance. [4]

2 EIGHT POSITIONS

Looking forward to the work of In2writing 2023, I offer then the following position statements:

Position 1: I agree with the argument of co-organizer Mina Lee, et al. [6], in the need for a holistic framework for evaluating

Table 1: Descriptive statistics for quantitative and qualitative indices of writing production

	Smart Compose Disabled (*n*=59)				Smart Compose Enabled (*n*=55)			
	Min.	Max.	Mean	SD	Min.	Max.	Mean	SD
Quantitative Indices								
Number of paragraphs	1	14	4.39	2.91	1	10	3.84	2.47
Number of sentences	11	60	29.66	11.14	14	77	28.96	10.88
Number of words	303	1,096	595.56	190.42	241	1,245	585.85	205.05
Number of words/sentence	13.21	32.27	21.00	4.50	12.61	42.00	21.08	6.01
Qualitative indices								
Syntactic simplicity	-1.57	1.06	-0.30	0.60	-2.02	1.12	-0.32	0.61
Syntactic complexity	2.09	9.00	4.39	1.31	2.25	7.56	4.16	1.38
Lexical diversity	.40	.79	.62	.08	.50	.79	.63	.07
Reading ease	39.94	82.14	67.02	8.09	37.72	80.68	65.62	9.09
Referential cohesion	-1.06	2.19	0.41	.71	-1.10	2.14	0.36	0.75

the impact of writing assistants on the human writing interaction. As Lee et al. argue through the articulation of HALIE, it will be important to expand non-interactive evaluation to capture the interactive process, the first-person subjective experience, and notions of preference beyond quality.

Position 2: This "first-person subjective experience" should be defined rhetorically, or it should engage (1) the writer's purpose; (2) the genre of the writing project (e.g., text message, email, letter, essay); (3) the technological context of the writing task (e.g., mobile platform, desktop computing), and (4) the category of writing technology engaged (e.g., writing assistant, writing generator). cf. [6]

Position 3: Further, the first-person experience can be informed by the development of AI-writing tool categories, developed around strategies common writing stages. Similar to the "spices" approach of Wordtune (Zhao, 2022) [10], and the co-author approach of offering an AI-powered writing collaborator (Lee, et al., 2022) [5], the In2writing community could develop aspirational categories for shaping AI writing generator outputs, based on a more developed understanding of writing processes.

Position 4: We need clearer definitions of writing assistant versus writing generator. Historically, AI is a field that has long articulated the role of "the assistant," or a technological manifestation of contextually relevant information to guide human decision making.[7] The difference between assisting humans in the writing process, as opposed to generating writing on behalf of humans, is a significant shift. I believe that this shift is driven by the race to develop feature sets at the technological level, but has outpaced the ability of most human writers to fully take advantage of writing suggestions.

Position 5: The latest versions of ChatGPT have demonstrated the potential for emotional reactions of human users.[9] Even sophisticated users, who understand that the writing outputs of ChatGPT do not represent sentience, cannot seem to resist deeply emotional reactions.[8] We need to partner explicitly with professionals from the mental health community to develop guidelines for human-computer interactions with writing generators.

Position 6: We need to provide better policy guidance for the regulation of writing generators, as their potential for disruption will encourage some to try to severely limit or curtail their application.[2]

Position 7: We need to attend to the educational requirements of AI-writing generators, as the significant shift in digital literacies they represent will require a more sophisticated understanding of their operational parameters from users in order to maintain a productive interaction. This will require engagement with professional organizations of writing teachers at P-16 levels. The proposed work in creating a taxonomy at the In2writing workshop presents a significant step toward that engagement.

Position 8: The evolution of Wikipedia provides an informative path for the In2writing community to consider. As a similar disruptive technology twenty years ago, Wikipedia was widely positioned as a threat to the creation and proliferation of accurate knowledge. Through the endurance of the Commons-based Peer Production model (Benkler, 2002) [3], it has prevailed to become a model for online collaboration. Understanding the evolution of Wikipedia as a collaborative writing platform could inform the evolution of the writing assistant research community.

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